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Modelling the South African Tender Process for Operational Improvement: A System Dynamics Approach

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Abstract

Public procurement in South Africa is a complex system where many parts, such as supplier performance, budgets, regulations, and administrative processes, interact and influence one another. These interactions often create bottlenecks in the tender process, and the effects build up over time. Traditional linear methods struggle to explain these patterns because they don't show how delays and feedback loops shape overall performance. This study uses ideas from System Dynamics together with a simple Discrete-Time model to examine how tender delays develop and spread through the procurement process. Using National Treasury data, the research focuses on understanding how long tenders take to move through different stages, why delays occur, and how changes in the process could improve performance. The model also tests how improvements like faster approvals or digital processing might reduce backlogs. By linking real data with a simple dynamic model, this study highlights the main sources of delay and shows how small changes can improve efficiency over time. The findings aim to support better planning, transparency, and decision-making in South Africa's public procurement system.

Keywords: System Dynamics, Procurement, Tender Process, Public Sector, Stock-Flow Modeling, Operational Improvement, Simulation

1. Declaration

I, **Dimpho Mola**, student number **65431006**, hereby declare that this research project titled *"Modelling the South African Tender Process for Operational Improvement: A System Dynamics Approach"* is my own original work and has not previously been submitted for any degree at this or any other institution.

In accordance with academic integrity guidelines, I further declare the following:

1. Use of AI Tools, Software, or Digital Platforms

I have used the following AI tools, software, and digital platforms during the course of this research:

Table 1: Tools and Platforms Used in this Study

Tool / Platform	**Purpose of Use**	**Specific Content or Section Supported**
ChatGPT (OpenAI)	Supporting conceptual understanding, clarifying complex ideas.	Simplifying procurement and system dynamics concepts for clearer comprehension.
Grammarly	Enhancing grammar, coherence, and academic tone.	Final language refinement across the full report.
Microsoft Excel, Vensim, Python	Data cleaning, statistical analysis, modeling, and visualization of procurement data.	All tables, charts, and simulation figures in the project.
TeXShop, LaTeX, and BibTeX	Document preparation, formatting, and citation management.	Full report structure, typesetting, and bibliography management.
Microsoft Word	Drafting preliminary sections and editing text collaboratively.	Initial report drafts before LaTeX typesetting.
Google Scholar and Google	Litrature Search and reference management	litrature referenced in this study was sourced from google scholar and google.

2. Purpose and Scope of AI Use

The AI tools listed above were used to assist in the research process, and not as a substitute for my own critical thinking, academic judgment, or analytical work. All AI-generated content has been carefully reviewed, verified, and edited by me to ensure

accuracy and academic integrity. Where applicable, original sources have been appropriately cited.

3. Acknowledgement of Responsibility

I accept full responsibility for the quality, accuracy, and integrity of this research project. I understand that any undisclosed or inappropriate use of AI tools may constitute academic misconduct and may result in disciplinary action in line with UNISA's policies.

	19 January 2026
Signature	Date

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2. Introduction

Strategic procurement plays a crucial role in operational efficiency and supply chain performance. It underpins the sourcing, contracting, and management of essential services and infrastructure, particularly in the public sector where procurement represents a significant share of government expenditure OECD [2024]. Yet, persistent inefficiencies continue to undermine procurement effectiveness. Delayed contract awards, inadequate planning, weak supplier vetting, and extended lead times frequently disrupt service delivery and increase costs Myeza [2018], Ambe and Badenhorst-Weiss [2012].

These challenges are compounded by systemic complexity. Procurement systems are not isolated processes but dynamic systems shaped by feedback loops, delays, and nonlinear relationships Sterman [2000]. Factors such as approval authority levels, contract complexity, compliance requirements, and supplier selection criteria interact with broader external influences such as political pressure, market instability, and policy changes. These interactions often produce unintended consequences, making procurement performance difficult to predict and manage He et al. [2023], Benington and Hartley [2009].

Recent research demonstrates that complexity theory and system dynamics provide powerful tools for understanding such behavior Sterman [2000], Mohd Yusoff [2018]. This study applies System Dynamics to simulate procurement performance on the public procurement tender process using empirical data from the South African National Treasury. By capturing feedback effects, delays, and policy interventions over time, System Dynamics enables the identification of high-leverage interventions to improve efficiency and reduce systemic delays in the public procurement process.

2.1. Research Overview

This research investigates strategic procurement performance challenges in South Africa's public sector, a domain characterized by long procurement cycles, irregular expenditure, and underperforming contracts despite continued investment in digital tools and policy reform Ambe and Badenhorst-Weiss [2012], Auditor-General South Africa [2023]. Existing studies highlight that these issues are not merely operational but result from structural and behavioral complexity within public procurement systems Myeza [2018], He et al. [2023].

Building on this foundation, the study applies System Dynamics to simulate how internal processes and external shocks interact over time. Unlike traditional static models, dynamic simulations offer a way to capture feedback loops, time delays, and nonlinear relationships that often drive unexpected outcomes in procurement systems Sterman [2000],

Mohd Yusoff [2018].

The model will be informed by publicly available data from the South African National Treasuery. Simulations will test scenarios including improved planning, supplier vetting, and digital platform implementation. The ultimate goal is to support strategic decision-making and contribute to procurement reform by providing evidence-based insights into system behavior as well as to identify high-leverage interventions that can improve procurement performance and inform more adaptive, policy aligned decision making.

2.2. Research Objectives

This study aims to:

- Describe historical tender activity by analysing trends in tenders published, tenders closed, and the resulting stock of active tenders over time.
- conceptualise the tender process as a dynamic feedback system using causal loop and stock–flow diagrams that capture delays, congestion, and process constraints.
- construct a discrete-time System Dynamics model that simulates the accumulation and clearance of active tenders.
- evaluate the impact of alternative process scenarios, such as improved onboarding efficiency and backlog-induced delays, on the long-run behaviour of the procurement system.
- Identify key leverage points within the tender process that can support more efficient and responsive procurement management.

2.3. Research Questions

1. How do feedback mechanisms, delays, and process efficiencies influence the accumulation and clearance of active tenders in the South African public procurement system?
2. How can System Dynamics be used to simulate tender processing performance over time?
3. Which factors most significantly affect procurement efficiency and backlog in the South African public sector?
4. How do selected process improvements (faster approvals, digital tools) impact procurement outcomes in the simulation?

3. Literature Review

Strategic procurement plays a vital role in attaining operational efficiency, especially in the public sector, OECD [2024] argue that procurement represents a significant share of public spending due to this being a critical part of public service delivery. Lagging indicators are often used in traditional performance assessments that often overlook the dynamic, feedback-driven nature of procurement systems. Maulanisa et al. [2024] continues to validate System Dynamics' applicability in procurement and project management contexts. For instance, Maulanisa et al. [2024]applies System Dynamics in engineering-procurement-construction projects to analyze delays and cash flow dynamics , while Liu et al. [2023] utilizes System Dynamics to model dynamic supplier selection incorporating Corporate Social Responsibility criteria, underscoring the method's versatility in complex supply systems. This literature review examines the application o System Dynamics in simulating procurement strategies in public sector entities. It additionally incorporates actual audit evidence from the Auditor-General of South Africa to emphasize real performance issues.

3.1. Procurement

3.1.1 Strategic Procurement in the Public Sector

Strategic procurement moves beyond cost minimization to encompass total value creation, supplier collaboration, risk management, and long-term strategic planning Matthews [2005]. Within the public sector, procurement processes are further influenced by legislative frameworks, political considerations, and social equity objectives, making transparency and accountability central to their effective execution Simone and Balasundharam [2023]. According to National Treasury, Republic of South Africa [2024], procurement performance is typically assessed using indicators such as lead time, procurement cycle time, supplier compliance, and cost efficiency.

Lead time refers to the duration between the initiation of a procurement requirement and the final delivery of goods or services. It reflects the responsiveness of the system and can be influenced by both internal administrative delays and external market factors. Procurement cycle time, on the other hand, measures the period between tender advertisement and contract award. This metric captures the efficiency of the procedural stages within the procurement process and is often extended by regulatory complexity, approval bottlenecks, and insufficient planning.

Supplier compliance evaluates the extent to which suppliers meet contractual, regulatory, and performance requirements. High compliance rates typically indicate effective supplier management and robust governance mechanisms, while low compliance may signal inadequate vetting or weak contract enforcement. Finally, cost efficiency assesses the

ability of the procurement system to achieve value for money while maintaining quality and timeliness. This involves not only minimizing direct costs but also managing hidden costs associated with delays, rework, and poor supplier performance.

While these metrics provide valuable insights into procurement performance, they often fail to capture the dynamic interdependencies within the system. Delays, supplier failures, and administrative inefficiencies interact over time through feedback mechanisms that conventional linear performance indicators are ill-equipped to represent. Understanding these nonlinear relationships is therefore essential for improving system-level efficiency and long-term strategic outcomes

3.1.2 Procurement Challenges in the Public Energy Sector

Public procurement systems, especially in large organizations, are not simple. They are made up of many parts that interact over time. These systems are constantly changing, depend on previous decisions, and include people and processes that react and adapt. Due to this, systems often behave in unexpected or even confusing ways OECD [2024].

Researchers such as Sterman [2000] explain that systems like these possess several defining characteristics. First, they are tightly connected, meaning that a small change in one part of the system can influence the entire structure. Ststems also contain feedback loops, where actions taken at one point can circle back to impact future decisions. These systems are often nonlinear, in that a minor action may result in a major effect, or, conversely, no noticeable effect at all. Furthermore, systems are adaptive, allowing people and processes within them to adjust in response to changing circumstances. Lastly, such systems typically involve delays, where the outcomes of current actions may not become evident until much later.These are all reasons why public procurement can be difficult to manage. For instance, Auditor-General South Africa [2023]highlights how delays in contract finalisation and supplier vetting at Eskom have contributed to emergency procurements and irregular spending, illustrating the real-world consequences of complex system interactions. An example we might look at can be that, speeding up approvals might cause quality issues later, or cutting budgets can lead to longer delivery times. These are what we call counterintuitive results, where solutions don't always work the way we expect Sterman [2000]. To better understand the dynamic and interconnected nature of procurement systems, Sterman [2000] advocates for the use of modelling tools such as System Dynamics and Discrete-Time Dynamical Models. These approaches enable the simulation of feedback relationships and time-dependent interactions among various elements of the procurement process. Through such models, researchers can examine how delays in supplier selection propagate through the system to affect downstream activities like delivery performance, or how repeated short-term, reactive decisions can accumulate into long-term structural inefficiencies. Furthermore, they allow for the exploration of

how financial constraints or budget reductions may introduce unintended risks, such as increased reliance on emergency procurement or compromised supplier quality. By capturing these feedback effects and time delays, System Dynamics provides a powerful lens for understanding how operational decisions shape long-term procurement performance.

Homer and Hirsch [2006] using System Dynamics in industries like healthcare show that this approach helps improve decision-making by revealing how the system actually behaves, not just how we think it does.

3.1.3 South African Tender Process

The Tender Process forms part of the crrutial sections of the South African Procurement System and an important part of the public service delivery. Tendering systems have been highlighted to have a range of structural, administrative and goverment challenges that affect the performance of the procurement system. An analysis of the south african goverment's tender system Ngobeni [2011] found that dispite compliance with legislation, the process is complex and often innieficent, whith practices that are outdated and inadeqate documentation hampering effective management. Procurement officials are often not provided with sufficient training which reflects mostly on the evaluation system which does not reflect best practices, further contributing to potential bias of contract awards. Roos and De La Harpe emphesises the importance of good goernance highlighting priciples such as independence, responsibility, accountability, efficiency and transparency within the public procurement sector, specifically in the tender process. Legal analyses note that public procurement in South Africa is regarded as an administrative process with significant implications for transparency and fairness, and that effective legal remedies and oversight are fundamental to upholding accountability and equitable outcomes. **?** also highlights the role of collusion and anti-competitive behaviour in the tender process, arguing that existing regulatory frameworks may not be robust enough to prevent bid-rigging or other forms of corruption. This underscores the need for stronger policy and institutional capacity to safeguard competitive and transparent tendering. **?** proposes reforms to public procurement law, including the development and refinement of a comprehensive national procurement statute, to simplify and improve the effectiveness, fairness, and competitiveness of procurement processes in South Africa. Such reforms aim to enhance transparency, reduce procedural delays, and strengthen oversight mechanisms.

3.1.4 Supplier Collaboration and Relational Dynamics

Procurement performance is not only influenced by structural variables but also by behavioral and relational factors. Boyce et al. [2016] used structural equation modeling to demonstrate that supplier collaboration, trust, and information sharing are critical to

improving procurement performance. These collaborative practices can be interpreted as reinforcing feedback loops in System Dynamics, where positive interactions among stakeholders lead to improved outcomes, which in turn foster further collaboration and trust. Walker et al. [2012]. These findings align with System Dynamics principles by emphasizing how relational dynamics act as reinforcing or balancing feedback loops that can either enhance or erode system stability.

3.2. System Dynamics

3.2.1 System Dynamics Fundamentals

System Dynamics provides a framework to study the nonlinear, feedback-driven behavior of complex systems over time. At its core, System Dynamics models are built on three interrelated components: stocks, flows, and feedback loops Sterman [2000].

Stocks represent accumulations in a system, such as the number of tenders awaiting adjudication, the pool of active suppliers, or allocated procurement budgets. They change over time through flows, which are the rates at which stocks increase or decrease. Mathematically, flows are expressed as ordinary differential equations that describe how a stock evolves over time Sterman [2000]. Due to most ordinary differential equations in real-world systems cannot be solved analytically, System Dynamics relies on numerical simulation methods, particularly the Euler method. This method approximates nonlinear continuous-time dynamics by breaking time into discrete slices and iteratively solving difference equations. The smaller the time step, the more accurate the approximation becomes. In applied settings, quarterly or annual steps are often used to align with decision-making cycles Meadows [2008]. A key strength of System Dynamics lies in its ability to capture feedback loops, closed chains of cause-and-effect relationships that drive system behavior. Reinforcing (positive) feedback accelerates growth or decline, such as when successful suppliers win more contracts, increasing their market dominance. Balancing (negative) feedback, in contrast, stabilizes a system by counteracting changes, for example when longer procurement delays discourage participation and reduce tender submissions. The interaction of these loops, combined with delays in decision-making or information flows, often produces oscillations, instability, or unintended outcomes in complex systems Otto et al. [2019]. To structure these relationships, System Dynamics uses Causal Loop Diagrams and Stock and Flow Diagrams. Causal Loop Diagrams provide a qualitative map of feedback relationships, while Stock and Flow Diagrams translate these into a formal structure for simulation. Together, these modeling tools allow policymakers to test alternative scenarios, such as digital procurement platforms or stricter supplier vetting, within a simulated environment before real-world implementation. In essence, System Dynamics equips researchers with both a conceptual language

and a computational toolkit to explore how procurement inefficiencies such as delays, irregular expenditure, and planning errors emerge from dynamic interactions over time. Linking real-world procurement data with System Dynamics models, it becomes possible to identify leverage points and simulate the potential impact of policy interventions.

3.2.2 Digitalization, Simulation, and Policy Testing

Simulation modeling is increasingly being used to test policy interventions, offering a structured and risk-free environment to anticipate how procurement reforms might perform before implementation. This approach enables decision-makers to explore multiple scenarios, evaluate the potential impact of interventions, and identify unintended consequences before real-world rollout. As shown in recent studies, simulation provides a powerful lens to assess complex systems such as healthcare, public infrastructure, and procurement, domains where feedback loops, delays, and adaptive behaviors are difficult to capture using traditional analytical methods Loblay et al. [2023]. South Africa's public energy sector, integrated system modeling, such as linking bottom-up energy systems models with macroeconomic general equilibrium frameworks, has been used to evaluate policy and investment scenarios in a simulated environment Altieri et al. [2019].

meanwhile, digital twin approaches enable real-time simulation of procurement-relevant outcomes (e.g., asset management, supply chain resilience), demonstrating how such models can guide interventions like platform integration, supplier audits, and infrastructure monitoring under dynamic conditions Madni and Madni [2019]. Zhang et al. [2021] argue that simulation techniques, whether system dynamics, discrete-event simulation, or hybrid approaches, allow policymakers to assess the long-term implications of strategic reforms within a controlled setting. Moreover, recent studies on digital twins highlight how integrating real-time data with simulation models can further enhance procurement system visibility, responsiveness, and agility. This supports the application of System Dynamics and Discrete-Time Dynamical Modelss in public sector procurement to simulate potential scenarios and identify robust policy responses to complex challenges such as supplier delays, irregular expenditure, and planning failuresZhang et al. [2021], Loblay et al. [2023].

3.3. System Dynamics in Procurement and Supply Chains

System Dynamics has been widely applied to procurement and supply chain systems to understand and improve performance under conditions of complexity and uncertainty. Mohd Yusoff [2018] demonstrated the value of System Dynamics in modeling procurement inefficiencies in the telecom sector, showing how feedback loops and delays result in unintended system behaviors. Mula et al. [2010] applied System Dynamics using Vensim

software to a multiperiod supply chain, highlighting its effectiveness in identifying bottlenecks and improving planning strategies. These studies indicate that System Dynamics is a suitable framework for public procurement modeling, particularly due to its capacity to incorporate feedback structures, time delays, and nonlinearity, which are commonly observed in complex procurement systems.

3.3.1 Feedback, Delays and Nonlinearity

The foundational work of Forrester [1961] in Industrial Dynamics laid the groundwork for applying System Dynamics in operational systems. His work illustrated how system inertia and poorly understood feedback mechanisms cause delays and oscillations in supply chains. Sterman [1989] and Lee et al. [1997] further explored the bullwhip effect, where minor demand changes upstream generate amplified fluctuations downstream. These works emphasize the importance of modeling dynamic interactions, which often result in unexpected or counterintuitive outcomes. Elmasry [2018] examined modular supply chain structures and demonstrated how internal system coupling impacts the emergent behavior of procurement networks. These insights justify the use of System Dynamics to uncover how nonlinear interactions, time delays, and feedback loops contribute to inefficiencies. Public procurement systems, especially in large organizations, are inherently complex. They consist of many interdependent parts that evolve over time, often influenced by past decisions and human behaviors. As Sterman [2000] explains, such systems exhibit the following characteristics: Procurement systems exhibit complex characteristics that make them challenging to manage using traditional linear models. These systems are defined by several dynamic features, including tight interconnections, where small changes in one component can trigger widespread impacts across the entire system; feedback loops, through which current actions influence future outcomes in either reinforcing or balancing ways; and nonlinearity, where seemingly minor interventions can lead to disproportionately large or negligible effects Sterman [2000]. Furthermore, procurement processes demonstrate adaptiveness, as stakeholders adjust their behaviors in response to evolving conditions, and delays, wherein the effects of decisions may only become apparent after significant time has passed, obscuring causal relationships. These dynamic properties often result in counterintuitive procurement behaviors, for example, expedited approvals intended to accelerate delivery may later lead to compromised quality, while short-term budget cuts can generate long-term inefficiencies and increased risk exposure. To analyze such systemic behaviors, researchers employ modelling approaches such as System Dynamics and Discrete-Time Dynamical Models, which simulate the temporal interactions among procurement variables. These models make it possible to explore how supplier selection delays influence delivery schedules, how reactive short-term decisions produce structural inefficiencies, and how financial constraints reshape procurement risk

profiles over time. By capturing feedback structures and time delays, these modelling approaches offer deeper insights into the underlying mechanisms driving public procurement performance.

Studies in manufacturing and healthcare sectors confirm the value of System Dynamics in revealing actual system behavior versus perceived behavior Coyle [1996], Katsaliaki and Mustafee [2011] This justifies the continued use of dynamic simulation to improve public procurement decision-making.

3.3.2 Insights from South African Public Sector Audits

The Auditor-General South Africa (AGSA) has consistently reported persistent inefficiencies within public procurement systems across national and provincial departments. In the 2022/23 Consolidated General Report, Auditor-General South Africa [2023] identifies several recurring challenges, including delays in finalizing contracts, poor planning and demand forecasting, inadequate supplier vetting and monitoring, and widespread instances of irregular, wasteful, and fruitless expenditure. These issues are problematic because they reflect systemic weaknesses that disrupt the procurement cycle at multiple stages. Delays in contract finalization, for example, often cascade into project implementation lags and increased costs, while poor demand forecasting undermines budget accuracy and resource allocation. Similarly, weak supplier vetting exposes departments to performance risks and non-compliance, leading to further administrative rework and financial losses. Irregular and wasteful expenditure erodes public trust and diverts funds away from critical service delivery. Collectively, these inefficiencies create reinforcing feedback loops, where delays cause cost escalations, which in turn lead to budget shortfalls and further delays, ultimately degrading overall system performance. Such dynamics underscore the need for analytical approaches like System dynamics which can capture these interdependencies and help simulate how small inefficiencies accumulate into larger systemic failures over time.

4. Methodology

This study followed a structured methodological approach to analyse and simulate the behaviour of the public sector tender process. Following the System Dynamics modelling framework proposed by Bajomo et al. [2022], a stock-flow representation of the tender process was developed with Active Tenders as the central stock.

Procurement data were sourced from the South African National Treasury's Open Tender Portal (2021-2025)National Treasury, Republic of South Africa [2024], comprising quarterly records of tenders published and tenders closed. A comprehensive data preparation stage was undertaken, including cleaning, filtering, aggregation into quarterly time series, and construction of the key flow variables required for modelling.

The System Dynamics model was then specified with Tenders Published as the inflow and Tenders Closed as the outflow. The cleaned data were incorporated to parameterise these flows over time. The model was simulated using a discrete-time formulation to generate baseline system behaviour and to test alternative policy scenarios, including improved onboarding and backlog pressure. Finally, the simulation outputs were analysed to identify structural drivers of delays, evaluate system responses under different conditions, and generate insights to support improved procurement performance.

4.1. Research Design

This research adopts a quantitative, simulation-based design grounded in System Dynamics, as articulated by Sterman [2000], and is implemented through discrete-time modelling. The tender process is inherently dynamic where tenders enter the system, accumulate as active cases, and exit once processing is completed. According to Albin [1997], such accumulations represent system stocks whose evolution over time is governed by inflows, outflows, and internal feedback structures. These dynamics respond to pressures such as workload intensity, administrative delays, and capacity constraints.

While traditional descriptive approaches can identify patterns in procurement data, Sterman [2000] argues that they are limited in their ability to explain how delays compound, how backlogs form, or how feedback between process stages shapes overall system performance. System Dynamics explicitly models these feedback mechanisms, allowing system behaviour to emerge endogenously from the interaction of stocks, flows, and time delays rather than being imposed exogenously.

To operationalise this perspective, the conceptual System Dynamics structure is translated into a discrete-time model aligned with quarterly tender data. This approach is consistent with the procurement focused System Dynamics framework proposed by Bajomo et al. [2022], which emphasises simulation as a tool for evaluating policy interventions in complex, delay-driven procurement systems. The stock of active tenders is updated iteratively based on observed system flows, enabling scenario simulations to examine the

effects of faster approvals, improved supplier onboarding, and congestion-induced delays.

Overall, this design provides a theoretically grounded and practically relevant framework for evaluating procurement performance in a setting characterised by complexity, interdependence, and feedback-dominated dynamics, as highlighted by Sterman [2000].

4.2. Casual Loop Dagram

The tender process in public procurement operates as a feedback-driven system in which delays, congestion, and corrective mechanisms interact over time. To capture these dynamics conceptually, a Causal Loop Diagram is used to represent the key feedback structures underlying tender processing performance. The Causal Loop Diagram serves as a qualitative foundation for the subsequent stock–flow model by identifying how variables interact and how inefficiencies may either escalate or stabilise.

In this study, the Causal Loop Diagram focuses explicitly on the tender process rather than the full procurement lifecycle. The central variable is Active Tenders, which represents the accumulation of tenders currently in progress. Changes in this stock are influenced by the rate at which tenders are published and the rate at which they are closed, with processing time acting as a critical mediating delay.

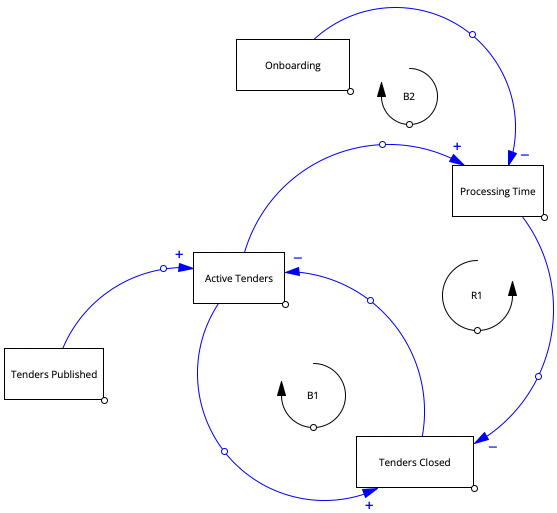

Figure 1: Tender Process Casual Loop Diagram

The dominant reinforcing mechanism in the system arises from backlog pressure. As more tenders are published, the number of active tenders increases. Higher congestion places strain on evaluation and approval capacity, which lengthens processing time. Longer processing time slows the rate at which tenders are closed, causing active tenders to accumulate further. This self-reinforcing loop explains how delays can compound even without external shocks.

Balancing mechanisms counteract this tendency. The primary balancing loop reflects process completion: as active tenders increase, administrative effort and throughput eventually rise, leading to more tenders being closed and reducing the backlog. A secondary balancing loop captures efficiency interventions such as improved onboarding procedures or digital tools. These interventions reduce processing time, accelerate tender closures, and relieve congestion.

The Causal Loop Diagram thus explains why procurement systems may experience persistent delays while also highlighting leverage points where targeted improvements can stabilise performance.

The tender process is characterised by one dominant reinforcing loop that amplifies congestion and two balancing loops that act to stabilise the system.

The tender process is governed by interacting reinforcing and balancing feedback loops that shape system behaviour over time. Table 2 summarises the key feedback mechanisms captured in the causal loop diagram.

Table 2: Reinforcing and Balancing Feedback Loops in the Tender Process

Loop ID	**Loop Name**	**Loop Type**	**Causal Structure and Interpretation**
R1	Backlog Pressure	Reinforcing (+)	An increase in tenders published raises the number of active tenders. Higher active tender levels increase processing time due to congestion, which slows tender closures. Slower closures further increase the number of active tenders, reinforcing backlog accumulation.
B1	Process Completion	Balancing (–)	As the number of active tenders increases, more tenders are processed and closed. Higher closure rates reduce the stock of active tenders, acting as a stabilising force within the system.
B2	Efficiency Intervention	Balancing (–)	Improvements in process efficiency, such as faster onboarding or digital tools, reduce processing time. Shorter processing times increase tender closures and lower the number of active tenders, alleviating congestion and stabilising system performance.

A "+" sign indicates that variables move in the same direction, while a "–" sign indicates movement in opposite directions.

4.3. Stock-Flow

The tender processing model is built around a small set of core variables that reflect how procurement activity evolves over time. The dependent variable in the model is Active Tenders, which represents the stock of tenders that have been published but are still moving through the evaluation, approval, and award stages. Its behaviour is directly shaped by the inflow and outflow of tenders in each quarter.

Figure 2: Tender Process Stock Flow: Core Flows

Figure **??** shows the primary control variables are the two flows entering and leaving the system. Tenders Published acts as the inflow and determines how many new tenders enter the active pipeline over time, while Tenders Closed serves as the outflow and represents tenders that have completed the process in a given quarter. Together, these two flows determine whether the stock of active tenders grows, stabilises, or declines.

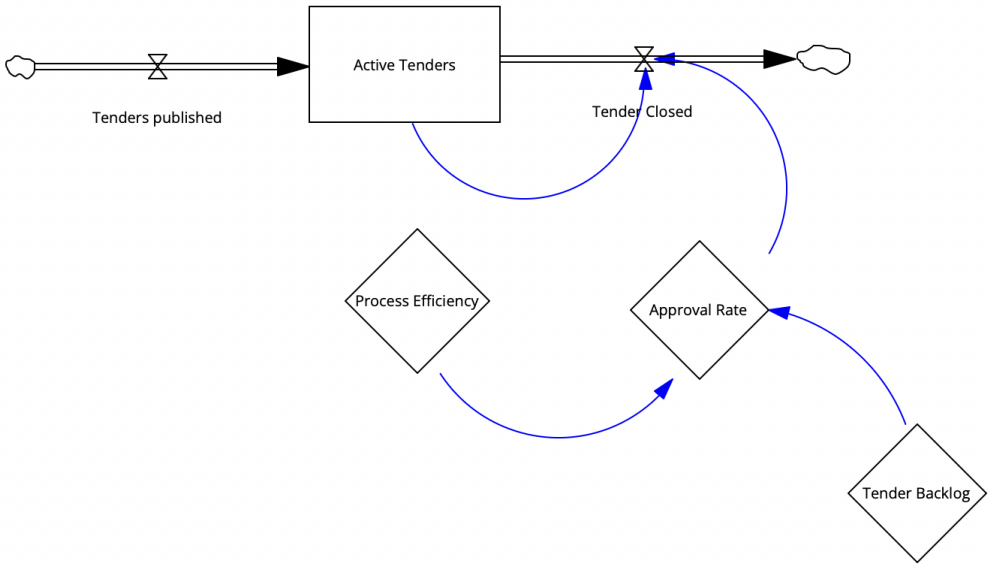

Figure 3: Tender Prodess Stock Flow: With Modifiers

In addition to the core flows, the model incorporates process-related modifiers that influence how efficiently tenders move through the system which is ilustrated in Figure **??**

. These include approval and evaluation delays, the efficiency of supplier onboarding, and backlog pressure effects. Although these modifiers do not appear directly in the National Treasury dataset, they function as internal parameters that shape how quickly tenders transition from published to closed. For example, an increase in backlog pressure slows processing because a larger volume of active tenders stretches administrative capacity, while improved onboarding accelerates closure rates by reducing administrative delays.

By structuring the model around these dependent, control, and process-modifying variables, the study captures the essential dynamics of the tender system without becoming unnecessarily complex. This allows the simulation to reflect both the observable behaviour in the historical data and the underlying mechanisms, such as delays and capacity constraints, that drive changes in procurement performance over time.

4.4. Stock-Flow Modeling of Active Tenders

In the System Dynamics model, Active Tenders is represented as the central stock variable, while Tenders Published and Tenders Closed function as its inflow and outflow, respectively.

Since our study focuses soley on the tender process, we can define a stock and flow diagram, which entails Relationship between stock and it's flow in the Stock-Flow diagram can be denotated as

$$AT(t) = AT(0) + \sum_{\tau=1}^{t} \left[\underbrace{TP(\tau)}_{\text{Inflow: Tenders Published}} - \underbrace{TC(\tau)}_{\text{Outflow: Tenders Closed}} \right]$$

Where in it's final form is written as:

$$AT(t) = AT(0) + \sum_{\tau=1}^{t} \big(TP(\tau) - TC(\tau)\big)$$

where $AT(t)$ denotes the stock of active tenders at time t, $TP(\tau)$ represents the inflow of tenders published at time τ, and $TC(\tau)$ represents the outflow of tenders closed at time τ.

The furmula explains that the change in the number of Active Tenders between time 0 and time t is equal to the accumulated difference between the tenders published and the tenders closed over that period. In other words, the stock of Active Tenders increases when new tenders are published and decreases when tenders are closed, and the net effect of these inflows and outflows over time determines the stock value at any given point.

In discrete time, consistent with the quarterly modelling framework of this study, the stock–flow relationship is expressed as:

$$\mathrm{AT}_t = \mathrm{AT}_{t-1} + x\mathrm{TP}_t - x\mathrm{TC}_t.$$

This means that changes in the stock between periods result entirely from the difference between the number of tenders published and the number of tenders closed in each quarter. The stock–flow formulation therefore provides a transparent representation of how procurement congestion, delays, or improvements propagate through the system over time.

For a simulation duration of T_{total}, calculations are performed at N equally spaced time points, separated by time steps of x. Starting from a given initial value of $Active\ Tenders(0)$, the stock variable at each time point i can be approximated iteratively by:

$$AT_t = AT_{t-1} + TP_t - TC_t$$

where t represents the efficiency factor applied to tender closures, and t denotes the discrete time step used in the simulation.

4.5. Data Collection

4.5.1 Data Sources

The data used in this study are sourced from publicly available South African National Treasury datasets, covering the period 2021–2025. These datasets were selected to support the study's objective of understanding tender processing dynamics and to parameterize a System Dynamics model focused on onboarding time and procurement cycle duration.

4.5.2 Core Datasets

The following datasets provide the empirical basis for the study and the simplified System Dynamics model:

- **Supplier Award.csv:** Captures awarded suppliers and identifiers, enabling mapping of outcomes at the supplier level.
- **Award.csv:** Provides award identifiers, supplier names, and monetary values, essential for analysing award distributions and contract sizes.
- **Main.csv:** Contains detailed procurement records, including tender ID, procuring entity, tender values, start and end dates, cycle times, award quarters and years, and metadata such as language, initiation type, and planning rationale.

These datasets allow the study to assess onboarding time as a central driver of procurement cycle duration, as highlighted in the causal loop diagram.

4.5.3 Optional Datasets

While not required for the current simplified System Dynamics model, the following datasets could be used in future research or extended models to capture additional feedback loops from the causal loop diagram:

- **tender_tenderers.csv:** Records supplier participation during tendering, enabling analysis of competitive intensity.
- **tender_documents.csv:** Metadata of supporting documents for assessing process transparency and potential bottlenecks.
- **parties.csv:** Links entities involved in tenders, supporting exploration of governance and stakeholder networks.

4.6. Data Preparation

Ensuring the reliability and consistency of the dataset is critical for both descriptive analysis and simulation modelling. The preparation process followed three key stages: cleaning, harmonisation, and transformation. Data cleaning involved resolving inconsistencies and improving dataset integrity. Duplicate records were identified and consolidated using unique award identifiers, ensuring accurate tracking of supplier-level activity over time. Missing or inconsistent entries for award dates, award values, and onboarding times were addressed either through imputation or removal, depending on the variable's importance. Textual inconsistencies in supplier names were corrected to ensure uniformity. Harmonisation focused on aligning datasets to facilitate integration and comparison. Column names were standardised across datasets, and all monetary values were converted into South African Rand (ZAR) to enable direct comparison. Date fields were reformatted into the standard YYYY-MM-DD structure, ensuring compatibility for temporal analysis and trend identification. Transformation prepared the cleaned and harmonised data for analysis and simulation. Key variables such as onboarding time, procurement cycle time, award values, and tender counts were aggregated by quarter. Procurement cycle time was computed as the difference between award date and request-for-quotation start date, formalised as: Additional derived metrics, including mismatch frequency and compliance rate, were generated to capture systemic dynamics identified in the causal loop diagram. These steps ensured that the dataset was consistent, complete, and suitable for the discrete-time modelling framework applied in this study.

4.6.1 Transformation and Integration

The raw procurement datasets obtained from the South African National Treasury were subjected to a systematic process of transformation and integration to ensure consistency

and usability for simulation. Date fields, including tender publication, closing, and award dates, were standardised into a uniform format to allow accurate temporal aggregation. Monetary values were converted to a consistent currency unit and aligned across datasets, while textual inconsistencies in supplier and entity names were harmonised. Key variables such as tender counts, onboarding times, and procurement cycle durations were aggregated at a quarterly level to match the discrete-time structure of the System Dynamics model. Derived metrics, including mismatch frequencies and compliance indicators, were created to capture systemic dynamics and administrative delays. Finally, the separate datasets were merged to form a unified analytical table, providing a clean, integrated foundation for modelling, simulation, and scenario testing.

4.7. Statistical Analysis

The analytical technique applied in this study centres on transforming quarterly procurement data into a dynamic representation of the tender process. The first step involved preparing and restructuring National Treasury tender records into a quarterly time series that captures the key flows entering and leaving the system. Once cleaned, the data were used to parameterise a discrete-time version of the System Dynamics stock–flow structure, allowing the tender process to be expressed as an accumulation equation that tracks how active tenders change from one quarter to the next. This formulation enabled the model to generate a simulated trajectory of the tender stock, which was compared to observed data to assess whether the structure adequately reflects real behaviour. After baseline validation, the model was extended to run scenario experiments. These scenarios modified specific components of the flow equations, such as the effective closure rate or the influence of backlog pressure to test how changes in process performance would propagate through the system over time. The simulations were then interpreted by examining how each scenario altered the level and stability of active tenders, providing insight into the operational consequences of proposed improvements. Through this combination of empirical time-series preparation, model calibration, and scenario testing, the analytical technique offers a systematic way to examine how tender-processing performance responds to delays, constraints, and policy adjustments.

4.8. Software and Tools

This study was conducted using a combination of analytical, statistical, and simulation-based software tools chosen for their suitability in handling time-series data, implementing system dynamics logic, and generating reproducible results.

Python was the primary environment for data cleaning, transformation, and modelling. Libraries such as pandas were used to structure the quarterly procurement data,

while matplotlib supported the creation of descriptive and simulation-based plots. Custom functions were written to operationalise the system dynamics equations in discrete time, enabling the simulation of tender accumulation under different scenarios.

RStudio was used during early exploratory work, particularly for validating quarterly patterns and ensuring consistency across data sources. Packages suited to time-series manipulation helped cross-check the integrity of the dataset before simulation.

LaTeX served as the documentation and reporting tool, ensuring coherence, mathematical clarity, and professional presentation. TikZ diagrams were used to construct a clean stock–flow visualisation of the tender process.

Vensim was used to build the stock–flow and causal loop diagrams, with Active Tenders as the central stock and Tenders Published and Tenders Closed as inflows and outflows. The software visualised key feedback mechanisms, such as backlog accumulation and compliance constraints, providing a clear representation of the procurement system. These diagrams guided the discrete-time simulations using quarterly National Treasury data, linking conceptual insights to quantitative analysis.

Together, these tools created a streamlined workflow from data acquisition, cleaning, and preparation, through modelling and simulation, to the final interpretation and reporting of results. The combination also ensures that the entire study is fully reproducible and transparent, with each step implemented in an open, script-based environment.

4.9. Scenario Testing and Policy Experiments

Scenario testing is used to evaluate how different tender processing conditions change the behaviour of the procurement system over time. Each scenario is based on the same stock–flow structure, where *Active Tenders* represents the accumulation of tenders currently in the system. The historical number of tenders published is kept constant, while the tender closure process is adjusted in each scenario.

The core stock equation in discrete form is:

$$AT_t = AT_{t-1} + TP_t - TC_t$$

where t represents a quarterly time step.

To calculate each quarter, the following step-by-step procedure is applied:

Step 1: Start with previous Active Tenders value

Step 2: (+)Add Tenders Published in quarter t

Step 3: (−)Subtract Tenders Closed in quarter t

This gives the new value of Active Tenders for quarter t.

4.10. Scenario Design and Simulation Logic

This study models the evolution of active tenders using a discrete-time stock–flow structure. At each quarter t, the number of active tenders is updated based on new tenders published and tenders closed during the period. The general stock–flow identity is given by:

$$AC_t = AC_{t-1} + TP_t - TC_t. \tag{4.1}$$

Three scenarios are constructed to analyse how different operational assumptions affect system behaviour over time.

4.10.1 Baseline Scenario

The baseline represents the observed system, where tenders are closed exactly as recorded historically. The quarterly update equations are:

$$TC_t^{(0)} = TC_t, \tag{4.2}$$

$$AT_t^{(0)} = AT_{t-1}^{(0)} + TP_t - TC_t. \tag{4.3}$$

This scenario reflects current procurement performance without any structural changes and serves as a reference point for comparison.

4.10.2 Scenario 1: Improved Processing Efficiency

This scenario represents reforms such as digital workflows or faster approval procedures that increase the system's processing efficiency. A constant efficiency factor $y > 1$ scales the number of tenders closed per quarter:

$$TC_t^{(1)} = y \cdot TC_t. \tag{4.4}$$

The corresponding stock update becomes:

$$AT_t^{(1)} = AT_{t-1}^{(1)} + TP_t - y \cdot TC_t. \tag{4.5}$$

An increase in y therefore reduces the accumulation of active tenders over time, illustrating how faster processing mitigates backlog growth.

4.10.3 Scenario 2: Backlog Pressure and Delay

This scenario captures congestion effects that arise when the number of active tenders exceeds processing capacity. A congestion factor x reduces the effective closure rate as

backlog increases, while K represents a normalisation constant linked to system capacity. The closure function is defined as:

$$TC_t^{(2)} = \frac{TC_t}{1 + x\left(\frac{AT_{t-1}}{K}\right)}. \tag{4.6}$$

The corresponding stock equation is:

$$ATs_t^{(2)} = AT_{t-1}^{(2)} + TP_t - \frac{TC_t}{1 + x\left(\frac{AT_{t-1}}{K}\right)}. \tag{4.7}$$

Under this formulation, a larger backlog increases the denominator of the closure function, reducing the number of tenders that can be processed. This creates a reinforcing feedback loop in which congestion leads to further congestion, reflecting real-world capacity constraints in public procurement systems.

4.10.4 Illustrative Calculation

To demonstrate how the stock–flow equations are implemented in practice, Table 3 presents an illustrative quarterly calculation of active tenders under the baseline and two alternative scenarios. The example shows how identical inflows can lead to different system trajectories depending on assumptions about processing efficiency and congestion effects.

Table 3: Illustrative scenario calculations for Active Tenders

Quarter (t)	**Active Tenders** ($t-1$)	**Published**	**Closed (Baseline)**	**Scenario 1:** $y = 1.2$	**Scenario 2:** $x = 0.5$, $K = 100$
2021 Q1	120	30	20	24 closed	18 closed
Results					
Baseline	$120 + 30 - 20 = 130$				
Scenario 1	$120 + 30 - 24 = 126$				
Scenario 2	$120 + 30 - 18 = 132$				

The parameters $K = 100$, $x = 0.5$, and $y = 1.2$ were selected to represent plausible operational conditions rather than to provide precise calibration. The threshold K reflects a moderate backlog level at which congestion effects become significant. The congestion scaling factor $x = 0.5$ represents a substantial but realistic reduction in effective processing capacity under overload, while the efficiency parameter $y = 1.2$ captures achievable improvements in closure performance arising from process optimisation. Together, these values enable transparent scenario comparison and highlight the sensitivity of the tender system to capacity and efficiency interventions.

Across all scenarios, the same accounting structure governs the evolution of the stock of active tenders. Differences in system performance arise solely from alternative assumptions about the tender closure process. This ensures that scenario comparisons reflect

genuine structural changes in system behaviour rather than differences in computational logic.

4.10.5 Comparative Evaluation

All scenarios are compared by plotting the trajectory of $ActiveTenders_t$ over the simulation period. The pattern of change indicates the system's behaviour:

- **Baseline** shows the current performance of the tender process and reflects how active tenders evolve under existing conditions.
- **Scenario 1** should display a downward trend in active tenders, demonstrating the impact of improved processing efficiency and faster closures.
- **Scenario 2** dislays a growth in active tenders, highlighting structural weaknesses caused by congestion and backlog pressure.

The comparison makes it possible to determine which intervention has the strongest effect and whether current processes are sustainable.

4.11. Tender Stock-Flow senario testing

Let $t = 0, 1, 2, \ldots, T$ denote discrete time periods measured in quarters. The initial condition $A\,T(0)$ represents the number of tenders already in progress at the start of the observation period.

At each period t, tenders enter the system through publication and exit the system once they are closed or awarded. The cumulative stock equation is defined as:

$$AT(T) = AT(0) + \sum_{t=1}^{t=T} \left[TP(\tau) - TC(\tau)\right].$$

This formulation states that the number of active tenders at time $t =$ T is equal to the initial number of active tenders plus the accumulated net difference between tenders published and tenders closed over all preceding periods.

For implementation within a discrete-time simulation framework, the model is expressed recursively. The stock of active tenders in period t is computed from the previous period as:

$$A\,T_t = A\,T_{t-1} + T\,P_t - T\,C_t.$$

This recursive form enables sequential calculation of the stock variable across quarters. For each time step, the inflow increases the stock while the outflow reduces it. When inflows exceed outflows, the number of active tenders increases, indicating congestion

or backlog accumulation. Conversely, when outflows exceed inflows, the stock declines, reflecting improved processing capacity or reduced demand.

To accommodate behavioural effects such as backlog pressure or process improvements, the outflow term may be adjusted using a functional modifier. The generalised form becomes:

$$AT_t = AT_{t-1} + TP_t - \phi(AT_{t-1}) \cdot TC_t,$$

where $\phi(\cdot)$ is a scaling function that captures nonlinear effects arising from congestion, administrative capacity, or efficiency interventions. This structure allows the model to represent reinforcing and balancing dynamics observed in real-world tender processing systems.

The stock–flow formulation provides a transparent and mathematically tractable representation of the tender process, forming the foundation for simulation, scenario testing, and subsequent analysis of prcurement process

4.11.1 Comparative Evaluation

Simulation results from the baseline and alternative scenarios are compared by analysing the trajectories of $ActiveTenders_t$ over the simulation horizon. Differences in trend direction, magnitude, and stability provide insight into the system's responsiveness to policy interventions. A sustained reduction in active tenders under Scenario 1 indicates potential efficiency gains, while divergence or amplification under Scenario 2 highlights structural vulnerability to backlog accumulation.

These scenario experiments demonstrate how small structural changes in tender processing can generate markedly different system behaviours, reinforcing the suitability of System Dynamics for evaluating procurement reform strategies.

5. Simulation Model

This simulation model directly addresses the research objective by operationalising a discrete-time system dynamics approach to capture the evolving dynamics of procurement performance in the South African public sector. Following the frameworks of Coyle [1996] and Sterman [2000], each quarter is treated as a distinct time step in which procurement variables are updated, enabling the model to capture both short-term fluctuations and cumulative effects over time.

Causal Loop Diagrams were developed to conceptualise the interdependencies among procurement delays, compliance rates, and supplier onboarding flows, in line with the principles of Senge [1990]. Procurement inefficiencies are represented as reinforcing and balancing feedback processes, reflecting the recursive nature of systemic delays, policy interventions, and performance improvements, as discussed by Thai [2001] and Walker and Brammer [2012].

The model is designed to reflect key performance indicators such as cycle time, compliance rates, and budget utilisation and to test potential optimisation pathways through scenario experimentation. This setup enables a systematic exploration of policy levers that could improve procurement efficiency and reduce systemic delays.

This modeling setup enables policy experimentation through simulation scenarios such as accelerated supplier sourcing and selection, stricter compliance enforcement, or reallocation of budget resources. The outputs, including quarterly cycle time trends and supplier award distributions were generated using Python and integrated into the broader data analysis to provide empirical grounding for system-level insights.

5.1. Data Analysis

This section presents the empirical and simulation-based analysis of the tender process. The analysis begins with descriptive time-series exploration of observed procurement data, followed by validation of the stock–flow formulation. Scenario simulations are then used to examine how changes in tender processing mechanisms affect system behaviour over time.

5.1.1 Baseline System Dynamics - Stock-Flow

Quarterly trends in tender activity were examined to assess patterns of inflow, closure, and backlog accumulation. Figure 4 presents the evolution of published, closed, and active tenders from Q2 2021 to Q3 2025.

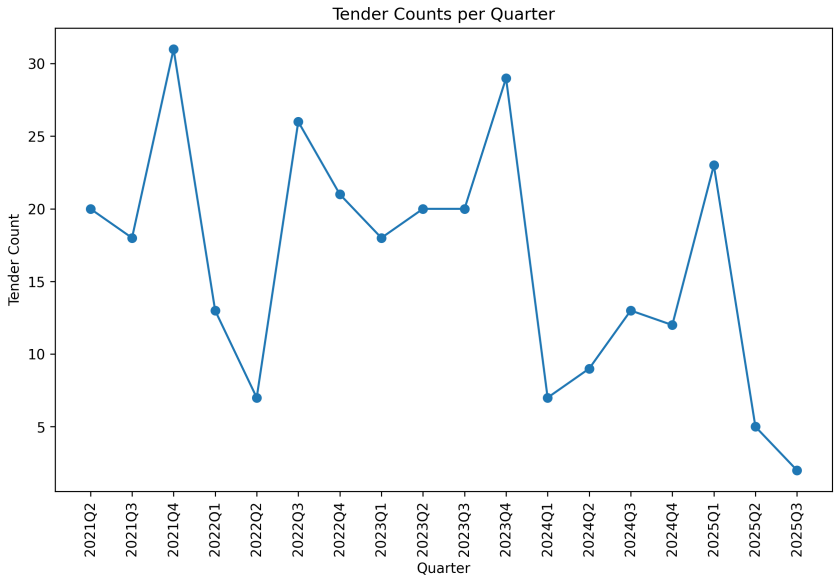

Figure 4: Quarterly tender activity: published, closed, and active tenders.

Interpretation: The number of active tenders exhibits pronounced fluctuations rather than convergence toward a stable equilibrium. Periods of backlog reduction are consistently followed by renewed accumulation, indicating that temporary improvements in throughput are not sustained. These cyclical patterns suggest that tender inflows frequently exceed processing capacity, generating recurring congestion. The sharp rebounds following short declines indicate a system operating near capacity limits, where modest increases in workload lead to disproportionate delays. Overall, the observed dynamics point to a structurally constrained procurement process characterised by recurring backlog formation rather than long term stabilisation.

5.1.2 Stock-Flow Construction and Validation

A discrete-time System Dynamics model was implemented to reconstruct the evolution of active tenders using quarterly procurement data. In the model, *Active Tenders* constitute the central stock, increasing through tender publication and decreasing through tender closure. This stock–flow formulation provides a parsimonious representation of the tender process and enables behavioural validation against observed data.

Three simulation scenarios were analysed: (i) a baseline scenario replicating historical inflows and outflows, (ii) a process improvement scenario representing faster tender processing, and (iii) a backlog-induced delay scenario capturing congestion effects under capacity constraints.

The baseline simulation closely reproduces the historical pattern of active tenders, providing behavioural validation of the stock–flow structure. The process improvement scenario produces a sustained decline in active tenders as increased closure rates improve system throughput. In contrast, the backlog scenario exhibits accelerating growth in

active tenders, reflecting reinforcing congestion effects whereby increased workload slows processing and further increases accumulation.

Table 4 summarises observed tender stocks, closures, and implied processing cycle times, illustrating the relationship between backlog levels and system throughput.

Table 4: Observed tender stock, closures, and simulated cycle time (2021–2025)

Period	Active Tenders	Tenders Closed	Cycle Time (Quarters)	Cycle Time (Days)
2021Q1	20	9	2.22	202.78
2021Q2	293	728	0.40	36.73
2021Q3	220	1002	0.22	20.03
2021Q4	239	1337	0.18	16.31
2022Q1	148	1671	0.09	8.08
2022Q2	414	991	0.42	38.12
2022Q3	450	2064	0.22	19.89
2022Q4	461	1788	0.26	23.53
2023Q1	514	1496	0.34	31.35
2023Q2	610	1655	0.37	33.63
2023Q3	705	2041	0.35	31.52
2023Q4	533	2265	0.24	21.47
2024Q1	510	1700	0.30	27.38
2024Q2	577	2095	0.28	25.13
2024Q3	679	2577	0.26	24.04
2024Q4	537	2322	0.23	21.10
2025Q1	627	2600	0.24	22.01
2025Q2	750	2741	0.27	24.97
2025Q3	29	1386	0.02	1.91
2025Q4	18	11	1.64	149.32

These results provide behavioural validation of the model and confirm that the tender process operates as a feedback-driven system in which small changes to processing mechanisms can generate substantially different outcomes over time.

5.1.3 Illustrative Scenario Calculations: 2025 Q1–Q4

To demonstrate the mechanics of the discrete-time stock-flow formulation, this subsection presents quarter-by-quarter calculations for the period 2025Q1-2025Q4. These calculations provide a transparent numerical link between observed procurement data and the scenario-based simulation results presented in the subsequent subsection.

The evolution of active tenders is governed by the standard stock–flow identity:

$$AT_t = AT_{t-1} + TP_t - TC_t, \tag{5.1}$$

where AT_t denotes the number of active tenders at quarter t, TP_t represents newly

published tenders, and TC_t denotes tender closures.

Three scenarios are evaluated:

- **Baseline:** observed tender closures.
- **Scenario 1 (Improved processing):** a proportional increase in closure capacity, modelled as $TC_t^{(S1)} = 1.2\,TC_t$.
- **Scenario 2 (Backlog-induced congestion):** reduced effective closure capacity when the backlog exceeds a threshold $K = 100$, modelled as $TC_t^{(S2)} = 0.5\,TC_t$ whenever $AT_{t-1} > K$.

The value $y = 1.2$ reflects a conservative 20% improvement in tender closure capacity, consistent with efficiency gains reported following process standardisation and digital procurement reforms in the public sector.

The congestion parameter $x = 0.5$ represents a substantial but plausible reduction in processing effectiveness under severe backlog conditions, reflecting administrative congestion, coordination delays, and capacity constraints observed in complex procurement environments.

The threshold $K = 100$ is used as a stylised representation of administrative capacity limits. While not a fixed operational constraint, it reflects a backlog level beyond which congestion effects become pronounced, as suggested by observed tender volumes and performance degradation in the dataset.

Table 5 reports the resulting active tender levels under each scenario. The observed value of active tenders in 2024Q4 ($AT_{2024Q4} = 537$) is used as the initial condition for 2025Q1.

Table 5: Illustrative stock-flow calculations for active tenders (2025Q1-2025Q4)

Quarter	Baseline	Scenario 1: Improved Processing	Scenario 2: Backlog Congestion
2025Q1	627	107	1927
2025Q2	750	202	2121
2025Q3	29	0	722
2025Q4	18	16	18

Table 5 illustrate how relatively small proportional changes in tender closure rates generate substantially different backlog trajectories. Enhanced processing capacity strengthens balancing feedback and rapidly reduces active tenders, while congestion effects activate reinforcing feedback that accelerates backlog accumulation when capacity thresholds are exceeded. These dynamics underpin the full simulation results discussed in the following subsection.

5.2. Sensitivity Analysis Results

Tables 6–8 present the results of the sensitivity analysis conducted on key structural parameters of the System Dynamics model, namely processing efficiency (y), congestion strength (x), and the capacity threshold (K). The analysis evaluates how variations in these parameters influence the evolution of the stock of active tenders over the period 2025Q1–2025Q4.

5.2.1 Processing Efficiency

Table 6 reports the sensitivity of active tenders to changes in processing efficiency (y).

Table 6: Sensitivity of Active Tenders to Processing Efficiency (y)

Efficiency (y)	**2025Q1**	**2025Q2**	**2025Q3**	**2025Q4**
1.0	627	750	29	18
1.1	367	216	0	0
1.2	107	0	0	0
1.3	0	0	0	0

The results indicate that increasing processing efficiency leads to a rapid reduction in the number of active tenders. This behaviour reflects a strong balancing feedback mechanism, whereby faster tender processing accelerates outflows and prevents backlog accumulation.

5.2.2 Congestion Strength

Table 7 shows the sensitivity of active tenders to congestion strength (x).

Table 7: Sensitivity of Active Tenders to Congestion Strength (x)

Congestion (x)	**2025Q1**	**2025Q2**	**2025Q3**	**2025Q4**
0.3	2447	4488	4737	4734
0.5	1927	3420	3392	3387
0.7	1407	2352	2047	2039

The results show that higher levels of congestion significantly increase the accumulation of active tenders. This pattern highlights reinforcing feedback effects, where congestion slows processing rates and further exacerbates backlog growth.

5.2.3 Capacity Threshold

Table 8 presents the sensitivity of active tenders to changes in the capacity threshold (K).

Table 8: Sensitivity of Active Tenders to Capacity Threshold (K)

Threshold (K)	**2025Q1**	**2025Q2**	**2025Q3**	**2025Q4**
50	627	750	29	18
100	627	750	29	18
200	627	750	29	18

The results indicate that variations in the capacity threshold have minimal impact on the evolution of active tenders over the simulated period. This suggests that, under the current workload conditions, the system does not reach levels where the capacity constraint becomes binding.

5.2.4 Scenario Testing: Simulation of Active Tenders

Scenario testing was undertaken to examine how changes in key tender process mechanisms influence the evolution of active tenders over time. The System Dynamics model represents *Active Tenders* as the central stock, increasing through tender publication and decreasing through tender closure. Using this structure, alternative scenarios were simulated and compared against the baseline trend observed in the historical data.

The first scenario represents improved supplier onboarding and faster internal processing. This reflects interventions such as streamlined evaluations, digital workflows, or reduced administrative delays. In the simulation, this improvement is implemented as a proportional increase in the tender closure rate, allowing tenders to exit the system more quickly.

The second scenario captures backlog-induced delays. As the number of active tenders grows, processing capacity becomes constrained, leading to slower evaluation and approval. This effect is modelled by reducing the effective closure rate as the stock of active tenders increases, introducing congestion-related delays into the system.

Figure 5 presents the simulated trajectories of active tenders under the baseline, improved onboarding, and backlog scenarios.

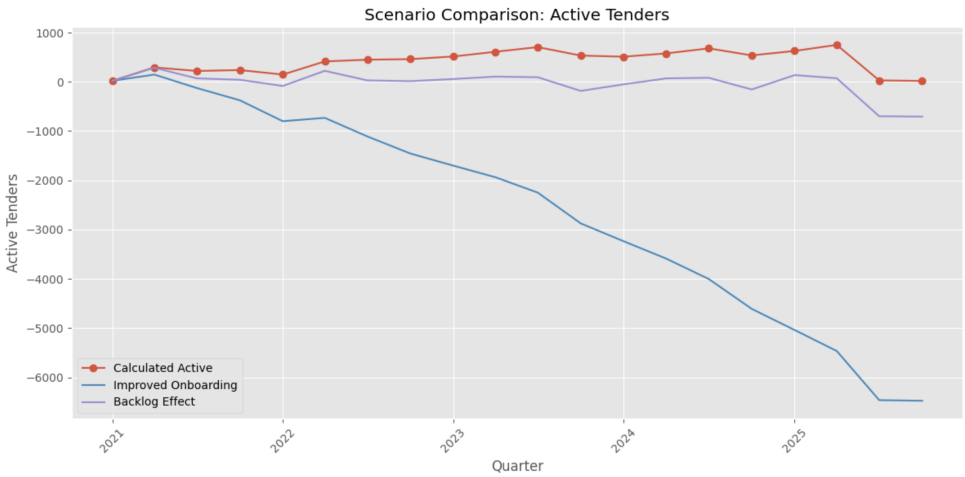

Figure 5: Simulation of active tenders under baseline, improved onboarding, and backlog scenarios.

Figure 5 show that improved onboarding leads to a sustained reduction in active tenders over time, indicating that faster processing alleviates congestion and improves system throughput. In contrast, the backlog scenario exhibits accelerated growth in active tenders, demonstrating how congestion can reinforce delays and increase accumulation when capacity constraints are present. Comparison with the baseline highlights the sensitivity of the tender system to relatively small changes in processing dynamics.

Overall, the scenario simulations reveal key leverage points within the tender process and confirm that procurement performance is shaped by feedback-driven dynamics. These findings demonstrate the value of System Dynamics as a tool for evaluating operational reforms and understanding how policy or process interventions may influence procurement outcomes over time.

5.2.5 Alignment of Causal Loop Diagram and Simulation Behaviour

The simulated behaviours align closely with the feedback structures identified in the Causal Loop Diagram, confirming the internal consistency of the System Dynamics framework. The baseline scenario reflects the interaction between balancing tender closures and reinforcing congestion pressures, resulting in persistent but bounded backlog levels.

The backlog-induced delay scenario activates the reinforcing loop associated with backlog pressure, whereby increasing active tenders strain processing capacity, lengthen cycle times, and further slow tender closures. This reinforcing dynamic produces accelerating backlog growth in the simulation.

Conversely, the improved onboarding scenario strengthens a balancing efficiency loop, increasing throughput and weakening congestion effects. As a result, active tenders decline steadily over time, indicating that efficiency gains can dominate reinforcing backlog dynamics when capacity constraints are alleviated.

This alignment between causal structure and simulated behaviour provides structural and behavioural validation of the model.

5.3. Discussion of Results

The simulation results demonstrate that procurement performance in the tender process is governed by feedback-driven dynamics rather than linear cause effect relationships. Under baseline conditions, balancing mechanisms are insufficient to offset sustained inflows, resulting in persistent backlog levels.

The backlog scenario highlights the vulnerability of the procurement system to congestion effects, where increases in workload generate self-reinforcing delays. In contrast, the improved onboarding scenario illustrates how targeted operational interventions can shift the system toward stronger balancing behaviour, reducing backlog accumulation without requiring reductions in tender volume.

These findings support the central premise of this study: procurement delays arise from endogenous system structure rather than external demand alone. System Dynamics provides a robust framework for identifying leverage points and evaluating policy interventions in complex public procurement systems. By integrating empirical data, feedback structures, and simulation outcomes, the analysis offers evidence based insights into improving procurement efficiency in the South African public sector.

5.4. Research Limitation

This study is subject to several limitations. First, the analysis relies on publicly available quarterly procurement data, which may contain inconsistencies or missing entries, potentially affecting the accuracy of model validation. Second, the System Dynamics model simplifies complex organizational processes through proportional closure rates and linear feedback loops, which may not capture all real-world decision-making processes. Finally, scenario simulations are hypothetical and parameter sensitive, limiting their predictive certainty. Future research could address these limitations by incorporating more data, and multi-level organizational decision-making.

6. Conclusion

This study has applied a System Dynamics perspective to analyse procurement performance in the South African public sector using tender data from the National Treasury for the period 2021–2025. The research progressed through a structured analytical process that included data cleaning, descriptive analysis, temporal alignment, stock–flow reconstruction, and scenario simulation. Together, these stages enabled a coherent understanding of both historical procurement patterns and the dynamic mechanisms that

shape system behaviour over time. Although the datasets used are publicly available, ethical research standards were upheld through transparency in data handling, transformation, and analytical procedures.

The findings demonstrate the feasibility and value of System Dynamics as an analytical framework for examining complex public procurement processes. By integrating empirical tender data with stock–flow simulations, the study shows how inefficiencies emerge from endogenous system structures, particularly delays and congestion within the tender process. Rather than viewing inefficiency as the result of isolated failures, the modelling approach reveals how feedback loops and accumulated backlogs systematically shape procurement outcomes. This perspective enables the identification of key drivers of inefficiency, supports the evaluation of alternative process configurations, and provides a foundation for generating evidence-based insights into operational improvement.

Beyond its analytical contribution, the study addresses a persistent and high-impact challenge in South Africa's public sector: inefficiencies in procurement that affect service delivery, fiscal discipline, and governance. Strategic procurement plays a critical role in ensuring that government departments can implement programmes effectively, manage public resources responsibly, and comply with regulatory frameworks. When tender processes are delayed or congested, the consequences extend beyond administrative inconvenience, increasing costs, weakening accountability, and slowing the delivery of essential services.

Despite ongoing reforms and digitalisation initiatives, many existing approaches to procurement analysis remain largely static and descriptive. Such methods are limited in their ability to capture the dynamic, feedback-driven nature of procurement systems. This research responds to that limitation by combining real-world National Treasury data with a simplified System Dynamics model of the tender process. By focusing on active tenders, inflows, outflows, and delays, and by simulating scenarios such as improved onboarding and backlog pressure, the study offers insights into system behaviour that traditional analytical tools cannot provide. The results demonstrate how small changes in processing efficiency or capacity constraints can generate markedly different long-term outcomes.

Looking forward, this modelling approach supports a more proactive and adaptive form of public procurement management. Rather than relying on trial-and-error implementation, policymakers and practitioners can use simulation to test potential reforms before they are introduced. In this way, System Dynamics becomes not only an analytical tool but also a strategic decision-support framework that can help identify high-leverage interventions, optimise tender workflows, and build resilience into procurement systems operating under fiscal and administrative constraints.

6.1. Recommendations

Based on the findings of this study, several recommendations are proposed. First, the System Dynamics model should be further refined and formally validated against historical tender flows to strengthen confidence in its representation of procurement dynamics. Second, a validated model can be used to test targeted policy and operational interventions, such as faster approvals, digital workflow implementation, and strategic management of tender inflows, to assess their potential impact on reducing systemic delays. Third, future extensions of the model should incorporate additional variables, including budget utilisation, contract values, and supplier performance indicators, to enhance its explanatory power and analytical depth. Finally, procurement institutions should consider adopting continuous monitoring mechanisms that track active tenders over time, enabling real-time feedback and supporting evidence-based decision-making.

In conclusion, this research provides a robust and adaptable framework for simulating and understanding strategic procurement performance in the South African public sector. By combining empirical data with System Dynamics modelling, the study offers a practical foundation for both analytical evaluation and policy experimentation. Most importantly, it equips public sector managers and policymakers with actionable insights to optimise tender management, reduce systemic delays, and improve operational outcomes in a context where efficiency, transparency, and accountability are essential to public trust and service delivery.

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